Land Value Uplift due to Mass Transit Development: Evidence from LRT-1

Thesis Proposal Presentation

Joanne Micaela Dizo Ma. Leibniz Charisse Parra Dianne Yumol

Department of Mathematics, Ateneo de Manila University

Table of contents

Introduction

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- **Traffic congestion** persists, emphasizing the need for improved connectivity [2].
- Proposed rail projects, costing **PHP 757 billion**, compete with budgets for health, education, and urban infrastructure [2].

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- Land value capture (LVC) utilizes land value increases from **improved accessibility** [2].
- It was successfully implemented in cities like **New York, London, and Copenhagen** [10].
- It remains underutilized in the Philippines despite demonstrated potential [8].

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- In Kuala Lumpur, HPM analyzed the effects of urban rail transit on property values, including proximity to stations and neighborhood factors [4].
- HPM and Difference-in-Difference models were also used to assess the **Dubai Metro's impact on property values**, accounting for accessibility and local economic factors [7].

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- GWR was also applied Tyne and Wear, UK, to explore spatially varying impacts of transit on property values [3].
- GWR accounts for **location-specific relationships**, offering detailed insights into spatial dependencies, complementing HPM for land value uplift analysis.

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- Many emerging cities lack comprehensive panel data, **making cross-sectional data more feasible**.

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- LRT-2 showed mixed residential results but significant commercial value increases, with a **Php 77.81 billion uplift** covering 46% of project costs [9].

This study focuses on evaluating the land value uplift induced by the Light Rail Transit Line 1 (LRT-1) on residential and commercial properties.

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- How can different property features affect its value?
- Does increased accessibility to transportation terminals and facilities lead to higher property values?
- What are the other significant factors (such as socio-economic factors) that influence property prices?

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- It captures a **snapshot of property valuations** around LRT-1 stations.
- It focuses on **spatial relationships** rather than tracking price changes over time.
- It is limited to properties marketed through **formal channels**, excluding those sold via traditional agencies or informal networks.

Data Collection

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- **72,019 unique listings** were extracted starting from August 2024 along the LRT-1 line.
- This data includes price, bedrooms, bathrooms, floor area, and land area.

Data Collection: Neighborhood Characteristics

Using QGIS with data sourced from OpenStreetMap (OSM), proximity data was derived and included the following:

1. Distance to the nearest LRT-1 station
2. Number of nearby hospitals, schools, malls, and bus stops

No distinctions were made among the different features; each was assigned an equal value of 1 point.

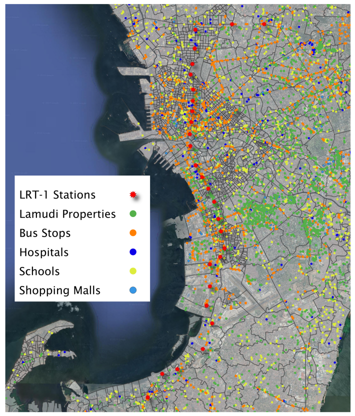

Figure 1: Data Collection through QGIS

Data Collection: Socio-Economic Data

Crime rate data

obtained via the Freedom of Information (FOI) Program to assess neighborhood safety.

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Socio-economic data has yet to be used for our preliminary models.

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4. No corresponding zonal value

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- Thus, if we are checking only residential properties, we remove outliers with respect to residential properties data alone.
- The same case is done for **normalization with min-max scaling** done after outlier removal, returning values from 0 to 1.

Data Preprocessing: Text Simplification

The property data includes a Description column which was cleaned of special characters and whitespace if in case this column be used in the future.

Preliminary Modeling

Model Testing: Variables

Dependent Variable: Property Price

Independent Variables: Property Characteristics:

- Number of bedrooms
- Number of bathrooms
- Floor area
- Land area

Independent Variables: Neighborhood Characteristics:

- Distance to the nearest LRT-1 station
- Number of nearby hospitals
- Number of nearby schools
- Number of nearby malls
- Number of nearby bus stops
- BIR zonal value

Model Evaluation: Variation Inflation Factor

Variation Inflation Factor (VIF)

$$VIF_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is R^2-value obtained by regression the j^{th} predictor on the remaining predictors.

Results: High multicollinearity is not evident among the independent variables.

Table 1: VIF Results for Independent Variables

Feature	VIF
Num_Bedrooms	1.890813
Num_Bathrooms	1.024081
Floor_Area	1.157715
Land_Area	1.000332
BusCount	3.866267
SchoolCount	3.117673
MallCount	2.081230
HospitalCount	1.969175
Matched_Zonal_Value	1.562212
LRTHubDist	1.393603

Model Testing: Hedonic Pricing Model

Hedonic Pricing Model (HPM)

$$y = \beta_0 + \sum_{i=1}^{n} \beta_i x_i + \varepsilon$$

where y is the property value, β_i is the regression coefficient, x_i are the independent variables, and ε is the error term.

- HPM estimates the contribution of individual property characteristics to its overall value.
- It breaks down the impact of each characteristic, capturing buyers' **"willingness-to-pay"** for each attribute.

Model Evaluation: Mean Square Error

Mean Square Error (MSE)

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2$$

where n is the number of data points, y_i are the original data collected, and $\widehat{y}_i$ are the predictions of the model being measured.

Model Evaluation: Adjusted R-squared

Coefficient of Determination

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \overline{y}_i)^2}$$

where n is the number of data points, y_i are the original data collected, $\widehat{y}_i$ are the predictions of the model being measured, and $\overline{y}_i$ is the mean of the original data.

Model Evaluation: Adjusted R-squared

Adjusted R-squared

$$\text{Adjusted } R^2 = 1 - \left(\frac{(1 - R^2)(n - 1)}{n - k - 1} \right)$$

where n is the number of data points, R^2 is the coefficient of determination, and k is the number of independent variables.

Results: The models generally perform better when the data is split into categories.

Table 2: Average R-squared and MSE values for 5-fold Cross Validated HPM

		#	Count		Distance	
			R^2	MSE	R^2	MSE
	General	25065	0.5511	0.0254	0.5562	0.0234
Property Type	RR	24714	0.5480	0.0262	0.5503	0.0249
	CR	1523	0.3025	0.0501	0.3229	0.0498
Category	House	14067	0.7065	0.0133	0.6975	0.0139
	Apartment	122	0.4125	0.0480	0.4068	0.0412
	Condo	12009	0.6624	0.0217	0.6674	0.0207
	Land	1508	0.2898	0.0542	0.3167	0.0532
	Commercial	118	0.5046	0.0423	0.5567	0.0433

Results: Floor area influences residential property price the most.

Table 3: HPM Results for Residential Properties

var	coef	std err	t	P>|t|
const	1.8e-16	0.004	4.19e-14	1.000
LRTHubDist	0.0098	0.005	2.052	0.040
Num_Bedrooms	-0.1490	0.009	-15.976	0.000
Num_Bathrooms	0.0680	0.009	7.458	0.000
Floor_Area	0.9487	0.009	103.879	0.000
Land_Area	-0.1972	0.008	-26.020	0.000
BusCount	0.0920	0.005	17.297	0.000
SchoolCount	-0.0144	0.005	-3.098	0.002
MallCount	0.1229	0.005	23.540	0.000
HospitalCount	-0.0924	0.005	-19.513	0.000
Matched_Zonal_Value	-0.0093	0.004	-2.076	0.038

Results: Land area and transport accessibility are valued in commercial properties.

Table 4: HPM Results for Commercial Properties

var	coef	std err	t	P>|t|
const	-1.83e-16	0.021	-8.52e-15	1.000
LRTHubDist	-0.2113	0.024	-8.684	0.000
Num_Bedrooms	-4.851e-18	6.89e-18	-0.704	0.481
Num_Bathrooms	-2.797e-17	1.39e-18	-20.111	0.000
Floor_Area	-4.509e-17	3.24e-18	-13.899	0.000
Land_Area	0.3246	0.022	14.598	0.000
BusCount	0.3486	0.024	14.755	0.000
SchoolCount	-0.0565	0.022	-2.612	0.009
MallCount	0	0	nan	nan
HospitalCount	0	0	nan	nan
Matched_Zonal_Value	-0.0951	0.022	-4.408	0.000

Results: Floor area is the most valued but number of bedrooms is not significant in house prices.

Table 5: HPM Results for Houses

var	coef	std err	t	P>|t|
const	2.458e-16	0.005	5.41e-14	1.000
LRTHubDist	-0.1095	0.005	-21.675	0.000
Num_Bedrooms	-0.0059	0.006	-0.987	0.324
Num_Bathrooms	0.0195	0.006	3.001	0.003
Floor_Area	0.6717	0.007	99.111	0.000
Land_Area	0.1900	0.006	32.541	0.000
BusCount	0.1576	0.005	30.230	0.000
SchoolCount	0.0200	0.005	3.782	0.000
MallCount	-0.0130	0.005	-2.725	0.006
HospitalCount	-0.0734	0.005	-14.194	0.000
Matched_Zonal_Value	-0.0955	0.005	-20.107	0.000

Results: Proximity to schools and hospitals is not significant for condominium prices.

Table 6: HPM Results for Condominiums

var	coef	std err	t	P>|t|
const	-2.279e-16	0.005	-4.26e-14	1.000
LRTHubDist	-0.0271	0.006	-4.245	0.000
Num_Bedrooms	-0.2791	0.008	-36.345	0.000
Num_Bathrooms	0.1382	0.007	19.660	0.000
Floor_Area	0.8817	0.007	118.366	0.000
Land_Area	3.833e-17	4.89e-18	7.842	0.000
BusCount	-0.0130	0.007	-1.837	0.066
SchoolCount	0.0011	0.006	0.185	0.853
MallCount	0.0373	0.007	5.456	0.000
HospitalCount	-0.0100	0.006	-1.575	0.115
Matched_Zonal_Value	-0.0321	0.005	-5.860	0.000

Results: Only the number of bedrooms, floor area, and land area are significantly affecting apartment price.

Table 7: HPM Results for Apartments

var	coef	std err	t	P>|t|
const	9.671e-17	0.073	1.32e-15	1.000
LRTHubDist	-0.0766	0.089	-0.859	0.392
Num_Bedrooms	0.4031	0.091	4.424	0.000
Num_Bathrooms	-0.0991	0.090	-1.098	0.274
Floor_Area	0.3525	0.077	4.563	0.000
Land_Area	6.411e-17	2.69e-17	2.383	0.019
BusCount	0.1098	0.089	1.239	0.218
SchoolCount	0.1147	0.081	1.419	0.159
MallCount	0.0146	0.078	0.187	0.852
HospitalCount	-0.1198	0.076	-1.568	0.120
Matched_Zonal_Value	-0.1080	0.075	-1.446	0.151

Results: Land area and transport accessibility are valued for land price.

Table 8: HPM Results for Land

var	coef	std err	t	P>|t|
const	-3.816e-17	0.022	-1.75e-15	1.000
LRTHubDist	-0.2128	0.025	-8.667	0.000
Num_Bedrooms	1.064e-18	7.13e-18	0.149	0.881
Num_Bathrooms	-4.238e-17	6.93e-18	-6.115	0.000
Floor_Area	2.436e-17	2.55e-18	9.557	0.000
Land_Area	0.3288	0.023	14.582	0.000
BusCount	0.3341	0.024	13.995	0.000
SchoolCount	-0.0627	0.022	-2.858	0.004
MallCount	0	0	nan	nan
HospitalCount	0	0	nan	nan
Matched_Zonal_Value	-0.0941	0.022	-4.298	0.000

Results: Relative to other models, commercial real estate properties values distance to the nearest LRT station the highest.

Table 9: HPM Results for Commercial Real Estate

var	coef	std err	t	P>|t|
const	1.284e-16	0.069	1.86e-15	1.000
LRTHubDist	-0.3407	0.090	-3.765	0.000
Num_Bedrooms	6.382e-17	4.13e-17	1.545	0.125
Num_Bathrooms	-3.974e-17	2.53e-17	-1.569	0.119
Floor_Area	0.3054	0.095	3.228	0.002
Land_Area	0.4649	0.091	5.095	0.000
BusCount	0.1567	0.084	1.877	0.063
SchoolCount	-0.2928	0.101	-2.898	0.005
MallCount	-0.1234	0.084	-1.477	0.143
HospitalCount	0.0284	0.096	0.295	0.768
Matched_Zonal_Value	0.0641	0.072	0.894	0.373

Model Testing: Localized HPM per LRT-1 Station

Figure 2: All properties from Lamudi

Figure 3: Properties within 2 km of stations

Results: Hedonic pricing models perform better when localized to a specific LRT station.

1. For the localized HPM per station, the range of R^2 values is from **0.219 to 0.976**.
2. The average R^2 is **0.715**.

A localized HPM model within a 2 km radius aligns with studies that typically define a reasonable walking distance to LRT stations as 1.6 km to 2 km [4, 6].

Results: Proximity to the nearest LRT-1 station significantly impacts the property prices of properties only near the Pedro Gil station.

Table 10: HPM Results for Properties within 2 km of Pedro Gil Station

var	coef	std err	t	P>|t|
const	0	0.040	0	1.000
LRTHubDist	0.1551	0.052	2.988	0.015
Num_Bedrooms	0.1557	0.064	2.444	0.037
Num_Bathrooms	1.203e-16	7.12e-17	1.690	0.125
Floor_Area	0.3712	0.073	5.113	0.001
Land_Area	-6.598e-17	7.08e-17	-0.932	0.376
BusCount	-0.0206	0.131	-0.158	0.878
SchoolCount	-0.5101	0.184	-2.777	0.021
MallCount	0	0	nan	nan
HospitalCount	0.1765	0.075	2.369	0.042
Matched_Zonal_Value	-0.0247	0.048	-0.512	0.621

Model Testing: LASSO Regression

LASSO (Least Absolute Shrinkage and Selection Operator) Regression

$$\widehat{\beta}_{\text{lasso}} = \underset{\beta}{\operatorname{argmin}} \left\{ \sum_{j=1}^{N} [y_i - (\widehat{\beta}_0 + \sum_{j=1}^{N} \widehat{\beta}_j x_j)]^2 + \lambda \sum_{j=1}^{N} |\beta_j| \right\}$$

where λ is the shrinkage parameter and $\sum_{j=1}^{N} |\beta_j|$ is the shrinkage penalty.

- LASSO Regression is a **regularization technique** that applies a penalty to prevent overfitting and improve model accuracy.
- It reduces errors caused by overfitting by adding a penalty term to the residual sum of squares (RSS), which is then multiplied by the shrinkage parameter (λ).

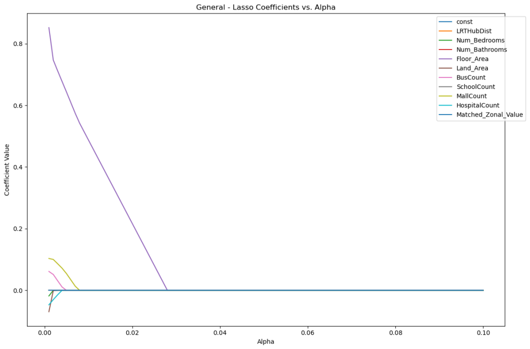
Results: LASSO Regression - General

Best Alpha: 0.0010

Cross-Validated R-squared (R^2): 0.5480

Cross-Validated Mean Squared Error (MSE): 0.0199

Feature	Coefficients
Floor_Area	0.0722
Num_Bedrooms	0.0016
MallCount	0.0016
Land_Area	0.0014
HospitalCount	0.0006
BusCount	0.0005
Num_Bathrooms	0.0001
SchoolCount	0.0000
Matched_Zonal_Value	0.0000
LRTHubDist	0.0000



Results: LASSO Regression - RR

Best Alpha: 0.0010

Cross-Validated R-squared (R^2): 0.5430

Cross-Validated Mean Squared Error (MSE): 0.0203

Feature	Coefficients
Floor_Area	0.0762
Land_Area	0.0030
MallCount	0.0014
Num_Bedrooms	0.0012
HospitalCount	0.0007
BusCount	0.0007
Num_Bathrooms	0.0002
SchoolCount	0.0000
Matched_Zonal_Value	0.0000
LRTHubDist	0.0000

Results: LASSO Regression - CR

Best Alpha: 0.0010

Cross-Validated R-squared (R^2): 0.2948

Cross-Validated Mean Squared Error (MSE): 0.0409

Feature	Coefficients
BusCount	0.0136
Land_Area	0.0119
LRTHubDist	0.0053
Matched_Zonal_Value	0.0010
SchoolCount	0.0004
Num_Bedrooms	0.0000
Num_Bathrooms	0.0000
Floor_Area	0.0000
MallCount	0.0000
HospitalCount	0.0000

Results: LASSO Regression - House

Best Alpha: 0.0010

Cross-Validated R-squared (R^2): 0.7096

Cross-Validated Mean Squared Error (MSE): 0.0117

Feature	Coefficients
Floor_Area	0.0370
Land_Area	0.0027
BusCount	0.0019
LRTHubDist	0.0009
Matched_Zonal_Value	0.0007
HospitalCount	0.0004
Num_Bathrooms	0.0000
SchoolCount	0.0000
MallCount	0.0000
Num_Bedrooms	0.0000

Results: LASSO Regression - Condo

Best Alpha: 0.0010

Cross-Validated R-squared (R^2): 0.6559

Cross-Validated Mean Squared Error (MSE): 0.0172

Feature	**Coefficients**
Floor_Area	0.0761
Num_Bedrooms	0.0071
Num_Bathrooms	0.0017
MallCount	0.0001
Matched_Zonal_Value	0.0001
LRTHubDist	0.0000
HospitalCount	0.0000
BusCount	0.0000
Land_Area	0.0000
SchoolCount	0.0000

Results: LASSO Regression - Apartment

Best Alpha: 0.0010

Cross-Validated R-squared (R^2): 0.2988

Cross-Validated Mean Squared Error (MSE): 0.0390

Feature	Coefficients
Num_Bedrooms	0.0163
Floor_Area	0.0141
HospitalCount	0.0017
SchoolCount	0.0016
BusCount	0.0014
Num_Bathrooms	0.0011
Matched_Zonal_Value	0.0009
LRTHubDist	0.0008
MallCount	0.0000
Land_Area	0.0000

Results: LASSO Regression - Land

Best Alpha: 0.0010

Cross-Validated R-squared (R^2): 0.2782

Cross-Validated Mean Squared Error (MSE): 0.0437

Feature	Coefficients
BusCount	0.0136
Land_Area	0.0127
LRTHubDist	0.0055
Matched_Zonal_Value	0.0010
SchoolCount	0.0005
Num_Bedrooms	0.0000
Num_Bathrooms	0.0000
Floor_Area	0.0000
MallCount	0.0000
HospitalCount	0.0000

Results: LASSO Regression - Commercial Real Estate

Best Alpha: 0.0010

Cross-Validated R-squared (R^2): 0.3539

Cross-Validated Mean Squared Error (MSE): 0.0362

Feature	Coefficients
Land_Area	0.0247
LRTHubDist	0.0128
Floor_Area	0.0106
SchoolCount	0.0083
BusCount	0.0022
MallCount	0.0015
Matched_Zonal_Value	0.0005
HospitalCount	0.0001
Num_Bedrooms	0.0000
Num_Bathrooms	0.0000

Model Testing: Geographically Weighted Regression

Geographically Weighted Regression (GWR)

$$y_j = \beta_0(u_j, v_j) + \sum_{i=1}^{n} \beta_i(u_j, v_j)x_{ij} + \varepsilon_j$$

where y_j is the property value at local point j, (u_j, v_j) are the location specific coefficients, x_{ij} are the independent variables, and ε_j is the error term for j.

- GWR accounts for the **spatial heterogeneity** by recognizing that the relationship between property values and independent variables can vary across different locations.
- *Modeling with GWR was attempted but it consistently encountered errors for most datasets, expect when limited to commercial real estate.*

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- HPMs are more effective at capturing pricing dynamics when they are applied at a **more granular level**.
- Across most models, **property characteristics** consistently have a significant influence on property price.
- The proximity to LRT-1 stations significantly impacts prices only for specific property types, namely:
    - Commercial Real Estate
    - Land
- However, when LASSO regression is applied, distance to the nearest LRT-1 station appears to be significant except for **condominiums**.

Plans Moving Forward

- Include socio-economic variables in the model
  - Crime rate
  - Network performance metrics
  - Family Income and Expenditure Survey data

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Appendix

Results: Distance to LRT has a significant negative effect on residential property price.

Table 11: HPM Results for Residential Properties with Neighborhood Distances

var	coef	std err	t	P>|t|
const	1.8e-16	0.004	4.21e-14	1.000
LRTHubDist	-0.0196	0.005	-4.322	0.000
Num_Bedrooms	-0.1419	0.009	-15.260	0.000
Num_Bathrooms	0.0731	0.009	8.055	0.000
Floor_Area	0.9664	0.009	107.018	0.000
Land_Area	-0.2338	0.007	-31.622	0.000
BusHubDist	-0.0815	0.005	-16.821	0.000
SchoolHubDist	0.0108	0.004	2.436	0.015
MallHubDist	-0.0934	0.005	-19.440	0.000
HospitalHubDist	0.1609	0.004	36.621	0.000
Matched_Zonal_Value	-0.0167	0.004	-3.783	0.000

Results: Distance to nearest bus stop has a negative effect on commercial property price.

Table 12: HPM Results for Commercial Properties with Neighborhood Distances

var	coef	std err	t	P>|t|
const	-1.83e-16	0.021	-8.63e-15	1.000
LRTHubDist	-0.1759	0.031	-5.757	0.000
Num_Bedrooms	-3.941e-17	1e-17	-3.940	0.000
Num_Bathrooms	1.142e-16	1.7e-17	6.724	0.000
Floor_Area	-1.444e-16	1.4e-17	-10.327	0.000
Land_Area	0.3516	0.023	15.600	0.000
BusHubDist	-0.3516	0.029	-12.150	0.000
SchoolHubDist	0.1619	0.024	6.734	0.000
MallHubDist	-0.0094	0.037	-0.253	0.800
HospitalHubDist	-0.1108	0.033	-3.406	0.001
Matched_Zonal_Value	-0.0899	0.021	-4.205	0.000

Results: Increased distance to both bus stop and LRT station have a negative impact on house price.

Table 13: HPM Results for Houses with Neighborhood Distances

var	coef	std err	t	P>|t|
const	2.458e-16	0.005	5.33e-14	1.000
LRTHubDist	-0.1195	0.005	-23.815	0.000
Num_Bedrooms	0.0026	0.006	0.428	0.669
Num_Bathrooms	0.0200	0.007	3.039	0.002
Floor_Area	0.6992	0.007	102.589	0.000
Land_Area	0.1692	0.006	28.893	0.000
BusHubDist	-0.1378	0.006	-22.291	0.000
SchoolHubDist	0.0228	0.005	4.521	0.000
MallHubDist	-0.0073	0.005	-1.337	0.181
HospitalHubDist	0.0571	0.006	9.546	0.000
Matched_Zonal_Value	-0.1140	0.005	-23.752	0.000

Results: Property characteristics mainly influences price of condominiums.

Table 14: HPM Results for Condominiums with Neighborhood Distances

var	coef	std err	t	P>|t|
const	-2.279e-16	0.005	-4.3e-14	1.000
LRTHubDist	-0.0311	0.006	-5.553	0.000
Num_Bedrooms	-0.2726	0.008	-35.748	0.000
Num_Bathrooms	0.1373	0.007	19.701	0.000
Floor_Area	0.8735	0.007	119.244	0.000
Land_Area	1.078e-17	6.27e-19	17.186	0.000
BusHubDist	0.0246	0.005	4.548	0.000
SchoolHubDist	0.0109	0.005	2.018	0.044
MallHubDist	0.0031	0.006	0.520	0.603
HospitalHubDist	0.0812	0.006	13.888	0.000
Matched_Zonal_Value	-0.0314	0.005	-5.785	0.000

Results: Among neighborhood characteristics, only the distance to the nearest hospital has a significant effect on apartment price.

Table 15: HPM Results for Apartments with Neighborhood Distances

var	coef	std err	t	P>|t|
const	9.671e-17	0.073	1.32e-15	1.000
LRTHubDist	-0.1516	0.081	-1.861	0.065
Num_Bedrooms	0.3742	0.093	4.038	0.000
Num_Bathrooms	-0.1187	0.091	-1.304	0.195
Floor_Area	0.3484	0.077	4.545	0.000
Land_Area	1.06e-16	3.29e-17	3.223	0.002
BusHubDist	-0.0961	0.086	-1.121	0.264
SchoolHubDist	0.0600	0.079	0.759	0.449
MallHubDist	-0.0738	0.079	-0.929	0.355
HospitalHubDist	0.1657	0.081	2.035	0.044
Matched_Zonal_Value	-0.1020	0.075	-1.357	0.178

Results: All neighborhood characteristics, except school, have an indirect relationship with land price.

Table 16: HPM Results for Land with Neighborhood Distances

var	coef	std err	t	P>|t|
const	-3.816e-17	0.021	-1.78e-15	1.000
LRTHubDist	-0.1669	0.031	-5.434	0.000
Num_Bedrooms	-1.305e-16	2.04e-17	-6.396	0.000
Num_Bathrooms	5.688e-17	1.05e-17	5.442	0.000
Floor_Area	2.731e-17	7.36e-18	3.711	0.000
Land_Area	0.3554	0.023	15.615	0.000
BusHubDist	-0.3467	0.029	-11.863	0.000
SchoolHubDist	0.1621	0.024	6.674	0.000
MallHubDist	-0.0120	0.037	-0.321	0.748
HospitalHubDist	-0.1168	0.033	-3.572	0.000
Matched_Zonal_Value	-0.0875	0.022	-4.059	0.000

Results: Proximity to bus stop is not significant but proximity to LRT station has a relatively high impact on commercial real estate price.

Table 17: HPM Results for Commercial Real Estate with Neighborhood Distances

var	coef	std err	t	P>|t|
const	1.284e-16	0.066	1.96e-15	1.000
LRTHubDist	-0.2757	0.082	-3.361	0.001
Num_Bedrooms	-3.464e-16	5.05e-17	-6.859	0.000
Num_Bathrooms	1.328e-16	3.35e-17	3.966	0.000
Floor_Area	0.2031	0.082	2.467	0.015
Land_Area	0.5281	0.086	6.132	0.000
BusHubDist	0.0081	0.073	0.111	0.912
SchoolHubDist	0.1862	0.074	2.532	0.013
MallHubDist	-0.3119	0.086	-3.609	0.000
HospitalHubDist	0.2496	0.082	3.056	0.003
Matched_Zonal_Value	0.1000	0.068	1.466	0.146

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Thank you!